

Notebook Guidelines

Tuesday, December 15, 2015 3:42 PM

Guidelines

Your notebook is your key for sharing the your ideas and work with your team, TA, and advisor. You can't get credit for your work and accomplishments if your advisor doesn't know about them. The expectation is that each one-credit student will work 3.5 hours per week outside of lab and each two-credit student will work at 5 hours outside of lab. The notebook should provide evidence of this level of effort.

Your notebooks will be evaluated by your TA throughout the semester. See the [Grading Guidelines](#) and the recommended content and format below for help acing your review.

For further guidance, see [EPICS Individual Notebook Guidelines](#)

You should contribute WEEKLY to each of the sections:

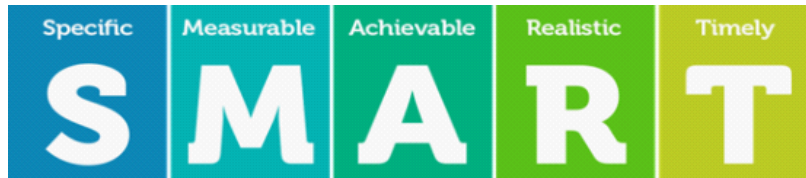
- **Action Items and Goals**
 - Starting with the end in mind and the overall project timeline, determine the proper GOALS for the semester and the critical path to achieve them.
 - Weekly: review the semester plan and goals, and meet to verify the team stays focused in the project goals considering the input for project partner, advisers, reviewers, and TA.
 - Identify the relevant action items, who is responsible for them, and proper due dates to maintain steady progress in the project.
- **Individual Work and Accomplishments**
 - Makes notes as you work. What are you doing? How is it relevant to the semester plan? How are you doing it? What observations have you made? How did you make decisions?
 - You should be working as a team, but describe here your specific contributions to that team and the weekly team's goals.
 - Include artifacts and text describing any brainstorming, sketches, calculations, designs, or action items.
- **Reflections**
 - Each week you will consider one of the four areas and critically reflect about what you have learned in that area this semester.
 - Use the guided questions to help you reflect.

Further suggestions for the notebook include:

- Lecture Notes
- Contact Information
- Conversations with project partner, consultants, advisors, or others
- Location of documents (SharePoint, etc.) or code (GitHub, SVN, etc.)
- Conclusions or recommendations
- Library or patent searches
- Research, including URL's for websites

Goals and Action Items

Wednesday, August 19, 2020 5:45 PM



- Starting with the end in mind and the overall project timeline, determine the proper GOALS for the semester and the critical path to achieve them.
- Weekly: review the semester plan and goals, and meet to verify the team stays focused in the project goals considering the input for project partner, advisers, reviewers, and TA.
- Identify the relevant action items, who is responsible for them, and proper due dates to maintain steady progress in the project.

WEEK 1

- Find group of interest
- Volunteered for a position with in the group
- Meet the people in my group
- Quick catch up on project

WEEK 2

- Start working on the box
- Started sanding
- Started filling screw holes with wood and glue

WEEK 3

- Continued filling holes with wood and glue
- Finished sanding
- Started painting the box

WEEK 4

- Finished painting the box
- Filled tube with water to check tube for leaks

WEEK 5

- Started sealing the box
-

WEEK 6

- Retested the tube for leaking
- Finished sealing the box with mega lock and tape
- Recheck airstones

WEEK 7

- Help and watch midterm review

WEEK 8

- Box still leaked so we tried using little bits of silicone sealant
- Added door latches
- Added top of the box latches

WEEK 9

- Started working on attaching wheels
- Retested the tube for leaks

WEEK 10

- Retested the tube for leaks
- Still leaked so we cleaned all megalock and tape
- Finished with the wheels

WEEK 11

- Epoxy on bottom of tube
- Replaced previous seal with plastic/silicone type of sealant for the tube
- Installed clamp for tube

WEEK 12

- Retested the tube for leaks
- Cut new PVC doors to shape

WEEK 13

- Clear coat all over box
- Paint new PVC doors
- Install new doors

WEEK 14

- Final seal test on tube
- Final design review

WEEK 15

- delivery

Work and Accomplishments

Tuesday, December 15, 2015 3:47 PM

Individual Work and Accomplishments Instructions

- **Makes notes as you work.** What are you doing? How are you doing it? What observations have you made? How did you make decisions?
- What did you get done? You should be working as a team, but **describe here your specific contributions** to that team.
- **Include artifacts** and text describing any brainstorming, sketches, calculations, designs, or action items.

Week 1 - 1/16/2026

As a first-time EPICS student, I began the workshop by listening to the different project groups before deciding to join the Bubble Tube team. I spent the beginning of the class getting familiar with the project and understanding where the team currently stands in the overall timeline. Returning members explained the progress made last semester and the tasks that still need to be completed. I also volunteered to serve as the group's project partner liaison and joined the iMessage group chat that was created for updates, photos, and coordination among team members. Later, the team went to the lab so that new members, including myself, could see the current state of the project and better understand the physical setup of the Bubble Tube system, which helped me visualize how the different components will eventually come together. For the remainder of the class, we brainstormed different ideas for integrating the electrical components into the box, since this was where the team had left off at the end of last semester. Finally, I completed the required lab safety quiz so that I can gain access to the lab and safely participate in future work on the project.

Week 2 - 1/23/2026

During this class, the team divided into two sub-teams, the electrical team and the box design team, to work more efficiently on different parts of the Bubble Tube project. I joined the box design team and contributed to brainstorming ideas for how the box should be designed. The discussion focused on how the box could securely hold the components while maintaining a clean and visually appealing design. We also considered how to leave enough internal space for the electrical components and other parts that will be installed later. In addition, the group created a Gantt chart to help organize the project timeline and outline the tasks our team plans to complete throughout the semester. It was then shared with the rest of the team so everyone could understand the overall plan and expectations moving forward. Lastly, I completed the required Personal PDH and the Safety quiz for the course.

[Quiz List](#) > [Submissions](#)

Quiz Submissions - Intro to EPICS Quiz (PDH#1) ▾

Individual Attempts	Grade
Attempt 1	5 / 8
Attempt 2	8 / 8
Overall Grade (highest attempt):	8 / 8

Summary - Purdue EPICS Safety Awareness (West Lafayette) ▾

Description

Click *Start Quiz* to begin.

Quiz Details

Current User

Spencer Lee (username: lee5035)

Time Limit

Unlimited

Availability

Due on Jan 16, 2026 11:59 PM

Available on Jan 5, 2026 12:01 AM

Attempts

Allowed - Unlimited, Completed - 1

Retake Incorrect Questions Only

You can only answer questions that were answered incorrectly on previous attempts.

All questions answered correctly

Week 3 - 1/30/2026

During week three, we focused on improving the finish of the Bubble Tube project box. Before sanding and painting, we filled in the previous screw holes using a mixture of wood glue and sawdust so the surface would be smoother and look more complete. After that dried, we sanded the box to remove rough spots and prepare it for paint. Once the sanding was finished, we painted the box so it would look cleaner and more polished. This step made the box look much more complete and closer to the final product. It was nice to see the project starting to come together physically as we worked through these finishing steps as a team. In addition to the work in the lab, I also finished the required Ethics Quiz for the course during this week.



[Quiz List](#) > [Submissions](#)

Quiz Submissions - Ethics Quiz (PDH#2) ▾

Individual Attempts	Grade
Attempt 1	5 / 5
Overall Grade (highest attempt): 5 / 5	

Week 4 - 2/06/2026

During week four, our team began the class by discussing the electrical work for the Bubble Tube project. As a group, we talked through how the electrical components might be implemented and what steps we would need to take moving forward. After the discussion, we shifted our focus back to the physical work on the box. We finished painting the box so that it looked more polished and complete. Completing the paint helped give the project a cleaner and more professional appearance. We also tested the tube by filling it with water to check for any leaks. This step was important to ensure the tube was properly sealed and would function correctly once the system is fully assembled. In addition to the work in lab, I also finished the required Personal PDH.



Design Exercise Quiz (PDH#3) ▾	Feedback: On Attempt	2 / 10
Due on Jan 30, 2026 11:59 PM		

Week 5 - 2/13/2026

During week five, our team focused on fixing the issue we identified during the previous leak test. We worked on sealing the tube more securely by applying tape around the areas where water might escape. The goal was to reinforce the seals and prevent any leaks before moving forward with the rest of the project. After sealing the tube, we filled it with water again to retest it and see if the additional sealing solved the problem. This step allowed us to check whether the tube would hold water properly and function as expected. In addition to the work in lab, I also finished the required Personal PDH.



Quiz Submissions - Design Process Module Quiz ▾

Individual Attempts	Grade
Attempt 1	4 / 4
Overall Grade (highest attempt):	4 / 4

Week 6 - 2/20/2026

During week six, we continued working on fixing the leak in the tube. The tape method we used previously did not fully solve the issue, as the tube was still leaking during testing. To address this, we decided to try sealing the tube again using Mega Loc to create a stronger seal around the problem areas. After applying the Mega Loc, we filled the tube with water and retested it with a full water pressure test to see if the new seal would prevent leaks. The Mega Loc helped reduce the leaking quite a bit compared to before, but the tube was still leaking slightly. Even though it was not completely fixed yet, this step helped us get closer to identifying and understanding what adjustments may still be needed. At the end of the class, we also rechecked the airstones to make sure they were functioning properly under pressure and producing bubbles as expected. In addition to the work in lab, I also finished the required Personal PDH.



Quiz Submissions - Design Process Guide Quiz (PDH#5) ▾

Individual Attempts	Grade
Attempt 1	4 / 5
Attempt 2	5 / 5
Overall Grade (highest attempt):	5 / 5

Week 7 - 2/27/2026

During week seven, we had our midterm review for the project. Although I did not actively present during the review, it was still very interesting to observe how the process worked. Since it was my first midterm review in EPICS, it was helpful to see how teams explain their projects and how reviewers evaluate the progress and design choices. Listening to the discussion gave me a better understanding of how our system works, especially when it came to explaining the pressure involved in the Bubble Tube setup. It was also valuable to hear the feedback from the reviewers. They pointed out a few areas that might become concerns later in the project, which gave our team things to think about as we continue improving the design.

Week 8 - 3/6/2026

During week eight, we focused on improving the functionality of the Bubble Tube project box. We attached door latches and additional latches for the top of the box to better secure the enclosure while still allowing easy access to the internal components when needed. This made the box feel more complete and practical to use. Overall, this step helped improve both the usability and overall quality of the design as we continue moving toward the final stages of the project.



Week 9 - 3/13/2026

During week nine, we began installing wheels on the box to improve mobility and make it easier to access the internal electrical components when needed. This addition helps make the system more practical to move and maintain. We also retested the tube after applying silicone sealant to the bottom section to improve the seal. This test helped us evaluate whether the new sealing method was more effective in preventing leaks and allowed us to continue refining the system's reliability.





Week 10 - 3/27/2026

During week ten, we retested the tube to check for any remaining leaks, but it was still leaking. Because of this, we decided to remove all of the previous sealing materials, including the Mega Loc and tape, so we could start over with a clean surface and try a different approach. This step was important because it allowed us to reset and better prepare for a more effective sealing method moving forward. In addition to working on the tube, we also finished installing the wheels on the box, which improved the mobility of the system and made it easier to access the internal components. Overall, this week focused on troubleshooting and improving both the functionality and practicality of the project as we continue refining the design.





Week 11 - 4/3/2026

During week eleven, we continued working on improving the seal of the tube to address the leaking issue. We noticed there was a crack at the bottom of the tube and fixed it by applying epoxy to create a stronger and more durable seal. In addition, we replaced the previous seal with a plastic/silicone-type sealant to improve overall sealing performance. We also installed a clamp to better secure the tube in place, helping maintain stability and reduce movement. Overall, this week focused on strengthening the structure of the tube and continuing to improve the reliability of the system as we move closer to a final solution.



Week 12 - 4/10/2026

During week twelve, we retested the tube again to check if the previous sealing improvements had reduced or eliminated the leaks. This step was important to evaluate whether our latest changes were effective and to identify if further adjustments were still needed. In addition to testing, we worked on modifying the structure of the box by cutting new PVC doors to the correct shape. This helped improve the fit and overall functionality of the enclosure. Overall, this week focused on continuing to refine both the sealing of the tube and the structural components of the project.

Week 13 - 4/17/2026

During week thirteen, we focused on finishing touches for the Bubble Tube project box. We applied a clear coat over the entire box to protect the surface and give it a smoother, more polished appearance. This step also helps improve durability by adding an extra layer of protection. In addition, we painted the new PVC doors that we had cut previously so they would match the rest of the box and maintain a consistent look. Overall, this week focused on improving the final appearance and durability of the project as we move closer to completion.



Week 14 - 4/24/2026

During week fourteen, we conducted the final seal test on the tube to determine whether all of our previous fixes had successfully resolved the leaking issue, which was important in confirming the system's reliability before completion. By this stage, we had applied multiple sealing methods and improvements, so this final test helped ensure the tube was functioning properly and ready for use. In addition, we participated in a design review for the project, and unlike the midterm review earlier in the semester where I mainly observed, this time I actively participated in presenting and discussing our design. This experience helped me better understand how to communicate our progress, explain design decisions, and respond to feedback. Overall, this week focused on both verifying the system's performance and improving my ability to engage in the design review process while incorporating feedback to finalize the project.

Weekly Reflections

Tuesday, December 15, 2015 3:42 PM

Weekly Reflections

Each week, you will consider one of the four themes of your choice below and critically reflect about what you have learned in that area this semester. Each reflection should include the following content components. The questions listed under each theme are for inspiration; they do not all need to be addressed or answered exactly as written.

Three Content Components

1. Reflective observation of your experience during the week to spark this reflection (Consider: identifying an experience and its context and/or impact).
2. Conceptualizing and connecting your observation to a broader concept in one of the reflection themes (Consider: What did I learn? How did I learn it? Why does the learning matter?)
3. Connecting how you will use your experience and learning in the future, inside and outside EPICS (Consider: What will/could I or others do in light of this learning?).

Four Themes:

a. Personal and Professional Development:

- i. What did I learn about who I am (my strengths, weaknesses, skills, etc.) and who I want to become, personally or professionally?
- ii. What assumptions or expectations did I bring to the situation? How did they affect what I did or didn't think, feel, decide, or do? To what extent did they prove true? If they did not prove true, why was there a discrepancy?
- iii. How did this situation challenge or reinforce my values, beliefs, convictions (e.g., my sense of right and wrong, my priorities, my judgments)? My sense of personal identity?
- iv. What is/should be the role of partnership, reciprocity, and/or co-design in my work? What have I learned from the community partner, and what resources have they provided to the project process and deliverable?
- v. What role does/should empathy play in my work?

b. Social Impact:

- i. What did I learn about the broader impacts of my work and how I and others can affect change locally and/or globally?
- ii. What did I learn about the community, our partners, the needs, and/or the quality of the service provided? What is the mission of our community partner and how do I support this through my effort in EPICS?
- iii. In what ways did differentials in power emerge in this experience? What are the sources of power in this situation?
- iv. In what ways is the individual good (mine or others) linked to and/or contrary to the common good? What trade-offs between them are involved? Who makes/made the trade-offs? Were the trade-offs made appropriate or inappropriate and why?
- v. In taking the actions I / they did, was the focus on symptoms of problems or causes of problems? Was the focus (symptom or cause) appropriate to the situation? How might I / they focus more on underlying causes in the future?

c. Academic Enhancement:

- i. What did I learn related to my discipline and how was that enhanced by the service-learning context? How can my discipline contribute in the context of community engaged design projects and impact society?
- ii. What skills, knowledge, and/or experience do I have that teammates might overlook, and how should this guide how we complete our project plan? What skills, knowledge, and/or experience gaps do I or my team have, and how might these be best addressed?
- iii. What did I learn about the design process and/or human-centered design?

- iv. What have I learned and what habits have I formed to improve my abilities and interest in lifelong learning?
- v. What is/should be the role of teamwork and inclusion in my work? How has my team been functioning and how can this be improved in the future?
- d. **Ethics:** (Consider referring to the [NSPE Code of Ethics for Engineers or other codes of ethics](#))
 - i. What have I learned about professional ethics and how decisions regarding ethical issues are made individually and as a team?
 - ii. What ethical issues may arise or have arisen in my project, how should ethical decisions be made in light of these issues, and how might this affect project identification and/or risk assessment analysis?
 - iii. What ethical and professional responsibilities should I consider in order to make more informed judgments, which consider the impact of solutions in global, economic, environmental, and societal contexts, when encountering similar situations in the future.

Week 1 - Personal and Professional Development

During my first EPICS workshop, I listened to several teams present their projects before deciding to join the Bubble Tube team. As a new EPICS student, I spent the beginning of the class learning about the project timeline and the progress made during the previous semester. Returning members explained what had already been completed and what tasks still needed to be done. I also volunteered to serve as the team's project partner liaison and joined the group chat so I could stay updated on communication and coordination. Later, we visited the lab to see the physical setup of the Bubble Tube system, which helped me better understand how the components would work together. This experience helped me realize that even as a new member, taking initiative and being willing to communicate can help a team function more effectively. In the future, I will try to continue being proactive in team environments by volunteering for responsibilities and contributing to communication and organization.

Week 2 – Academic Enhancement

During this week, our team divided into two groups so we could focus on different parts of the Bubble Tube project more efficiently. One group focused on the electrical work while the other focused on the box design. During our discussion, we talked about the current progress of the project and what tasks we should begin working on next. This conversation helped us decide how to move forward and what priorities we should focus on first. As a team, we also created a Gantt chart to outline our timeline for the semester and organize the tasks that need to be completed. Having this timeline helped everyone understand the overall plan and how different parts of the project connect. Through this experience, I learned that planning and communication are important parts of the engineering design process. In the

future, I will continue to value team discussions and pay attention to structured planning when working on collaborative projects.

Week 3 – Academic Enhancement

During week three, we worked on improving the finish of the project box. Before sanding and painting, we filled the previous screw holes using a mixture of wood glue and sawdust so the surface would be smoother. After that dried, we sanded the box and painted it so it would look more polished and complete. Although these steps may seem small, they helped improve the overall quality and appearance of the box. This experience showed me that attention to detail is important in engineering projects because small improvements can make a big difference in the final result. In the future, I will try to approach projects with more patience and attention to detail so the final product is both functional and well finished.

Week 4 – Social Impact

During week four, we began class by discussing the electrical components of the Bubble Tube project and how they will be integrated into the system. After the discussion, we continued working on the box and finished painting it so that it looked more polished. We also filled the tube with water to test whether there were any leaks in the system. While this task was mainly technical, it reminded me that the reliability of a system is important when creating something that will be used by others. If a system leaks or fails, it affects the quality of the service or experience being provided. Through this experience, I realized that engineering projects are not just about building something that works, but also about making sure it functions reliably for the people who will use or benefit from it. In the future, I will try to think more about how the quality and reliability of designs can affect the people or communities they serve.

Week 5 – Academic Enhancement

During week five, our team worked on addressing the leak we found during the previous test. We sealed the tube using tape around the areas where water might escape and then filled the tube with water again to retest it. This allowed us to check whether the sealing method was effective. Through this experience, I learned more about the iterative nature of engineering design. Solutions often need to be tested and improved multiple times before they work properly. The process of sealing and retesting the tube showed me that experimentation and adjustment are normal parts of the design process. In the future, I will approach challenges with the understanding that refining a solution through multiple attempts is often necessary.

Week 6 – Ethics

During week six, we realized that the tape method did not completely stop the leak, so we applied Mega Loc to create a stronger seal. After applying it, we filled the tube with water and retested it. The leaking improved significantly, although a small leak was still present. We also rechecked the airstones to make sure they were functioning properly under pressure and producing bubbles as expected. This situation helped me think about the ethical responsibility engineers have to ensure that systems are reliable and safe. Even though the leak improved, it was important for us to recognize that the issue was not completely solved. In the future, I will remember that engineers must carefully test and evaluate their work before considering it finished.

Week 7 – Personal and Professional Development

During week seven, we had our EPICS midterm review. Although I did not personally present, it was still very interesting to observe the process. Since it was my first midterm review, I learned how teams explain their projects and how reviewers ask questions about design decisions and potential concerns. Hearing the discussion about our system, including how pressure affects the Bubble Tube setup, helped me better understand the project. The feedback from the reviewers also highlighted areas our team may need to improve. This experience showed me how valuable feedback can be in strengthening a project. In the future, I will try to actively seek and apply feedback, since outside perspectives can help improve both designs and teamwork.

Week 8 – Academic Enhancement

During week eight, we attached door latches and top latches to the box to secure the enclosure while still allowing access to the internal components. This hands on work helped me better understand how small mechanical additions can significantly improve functionality and usability in a design. I learned that even simple components like latches play an important role in user interaction and maintenance, which connects to the broader concept of practical design in engineering. It showed me that design is not just about building something that works, but also about making it easy to use and maintain. In the future, I will pay more attention to these smaller functional details when working on projects.

Week 9 – Social Impact

During week nine, we installed wheels on the box to improve mobility and accessibility, especially for accessing the internal electronics. We also retested the tube after applying silicone sealant to improve the seal. This experience made me think about how design choices affect the people who will interact with the product. Adding wheels may seem like a small change, but it makes the system easier to move and more accessible, especially in a space where flexibility is needed. I learned that engineering decisions can directly impact usability and inclusivity, which connects to the broader idea of designing with the user in mind.

Week 10 – Academic Enhancement

During week ten, we retested the tube for leaks, but it was still leaking. Because of this, we removed all previous sealing materials, including Mega Loc and tape, so we could start over with a clean surface. We

also finished installing the wheels on the box. This experience reinforced the importance of iteration in the design process. I learned that sometimes the best step forward is to restart and approach a problem differently rather than continuing to build on ineffective solutions. This connects to the engineering concept of iterative problem-solving, where testing and refining are essential.

Week 11 – Ethics

During week eleven, we applied epoxy to fix a crack at the bottom of the tube and replaced the previous seal with a plastic/silicone-type sealant. We also installed a clamp to better secure the tube. This week made me think about the ethical responsibility engineers have to ensure that systems are safe and reliable. Even though the project is for children, or especially because of that, it is important that the structure is stable and secure. I learned that engineers must carefully consider safety and durability when making design decisions.

Week 12 – Academic Enhancement

During week twelve, we retested the tube again to evaluate whether our sealing improvements were effective. We also cut new PVC doors to better fit the box. This experience helped me understand how precision and accuracy play a role in both functionality and design. Poorly fitted components can lead to inefficiencies or problems, while well-designed parts improve the overall system. I learned that engineering often involves refining and adjusting components to achieve the best fit and performance.

Week 13 – Personal and Professional Development

During week thirteen, we applied a clear coat over the box and painted the new PVC doors to match the overall design. This week was focused more on finishing touches, but it helped me reflect on my growth throughout the project. I realized that I had become more comfortable working with tools, contributing ideas, and collaborating with my team. At the beginning of the semester, I was less confident in hands-on tasks, but through repeated practice, I improved my skills. This connects to personal development and building confidence through experience. In the future, I will continue to challenge myself to step outside my comfort zone and develop new skills, especially in hands-on and team-based environments.

Week 14 – Personal and Professional Development

During week fourteen, we conducted the final seal test on the tube to confirm that the system was functioning properly. We also participated in a design review, and unlike the midterm review where I mainly observed, I actively contributed to presenting and discussing our project. This experience helped me improve my communication skills and confidence in explaining technical work. I learned that being able to clearly present ideas and respond to feedback is just as important as building the system itself. This connects to professional development, as communication is a key skill in both engineering and business environments. In the future, I will continue working on my ability to present ideas clearly and confidently, and I will actively engage in discussions and feedback opportunities.

Semester Reflection

Tuesday, August 18, 2020 1:40 PM

Semester Reflection

For two of the themes below, critically reflect about what you have learned in that area this semester. Each reflection theme should include the following content components and contain at least 300 words. The questions listed under each theme are for inspiration; they do not all need to be addressed or answered exactly as written.

Three Content Components

1. Reflective observation of your experiences during the semester to spark this reflection
2. Conceptualizing and connecting your observations broader concepts in one of the reflection themes
3. Connecting how you will use your experience and learning in the future, inside and outside EPICS

Themes (Choose Two):

1. **Personal and Professional Development:**
 - a. What did I learn about who I am (my strengths, weaknesses, skills, etc.) and who I want to become, personally or professionally?
 - b. What assumptions or expectations did I bring to the situation? How did they affect what I did or didn't think, feel, decide, or do? To what extent did they prove true? If they did not prove true, why was there a discrepancy?
 - c. How did this situation challenge or reinforce my values, beliefs, convictions (e.g., my sense of right and wrong, my priorities, my judgments)? My sense of personal identity?
 - d. What is/should be the role of partnership, reciprocity, and/or co-design in my work? What have I learned from the community partner, and what resources have they provided to the project process and deliverable?
 - e. What role does/should empathy play in my work?
2. **Social Impact:**
 - a. What did I learn about the broader impacts of my work and how I and others can affect change locally and/or globally?
 - b. What did I learn about the community, our partners, the needs, and/or the quality of the service provided? What is the mission of our community partner and how do I support this through my effort in EPICS?
 - c. In what ways did differentials in power emerge in this experience? What are the sources of power in this situation?
 - d. In what ways is the individual good (mine or others) linked to and/or contrary to the common good? What trade-offs between them are involved? Who makes/made the trade-offs? Were the trade-offs made appropriate or inappropriate and why?
 - e. In taking the actions I / they did, was the focus on symptoms of problems or causes of problems? Was the focus (symptom or cause) appropriate to the situation? How might I / they focus more on underlying causes in the future?
3. **Academic Enhancement:**
 - a. What did I learn related to my discipline and how was that enhanced by the service-learning context? How can my discipline contribute in the context of community engaged design projects and impact society?
 - b. What skills, knowledge, and/or experience do I have that teammates might overlook, and how should this guide how we complete our project plan? What skills,

- knowledge, and/or experience gaps do I or my team have, and how might these be best addressed?
- c. What did I learn about the design process and/or human-centered design?
 - d. What have I learned and what habits have I formed to improve my abilities and interest in lifelong learning?
 - e. What is/should be the role of teamwork and inclusion in my work? How has my team been functioning and how can this be improved in the future?
4. **Ethics:** (Consider referring to the [NSPE Code of Ethics for Engineers or other codes of ethics](#))
- a. What have I learned about professional ethics and how decisions regarding ethical issues are made individually and as a team?
 - b. What ethical issues may arise or have arisen in my project, how should ethical decisions be made in light of these issues, and how might this affect project identification and/or risk assessment analysis?
 - c. What ethical and professional responsibilities should I consider in order to make more informed judgments, which consider the impact of solutions in global, economic, environmental, and societal contexts, when encountering similar situations in the future.

Personal and Professional Development

Throughout this semester in EPICS, I learned a lot about how I work in a team and how I handle new situations. At the beginning, I expected to mostly observe since I was a first time EPICS student and didn't have much experience with the project. However, that changed pretty quickly when I volunteered to be the project partner liaison and started getting more involved in discussions and hands-on work. As the weeks went on, I contributed to brainstorming ideas, working on building the box, and helping fix issues like the tube leaking. These experiences helped me become more comfortable sharing my ideas and taking initiative. One moment that really showed my growth was comparing the midterm review to the final design review. During the midterm, I mostly listened and observed, but by the final review, I was able to participate, explain parts of our design, and respond to feedback.

This connects to professional development because I learned that being a good team member is not just about doing assigned tasks, but also about speaking up, communicating, and being open to learning. At first, I think I held myself back a little because I wasn't fully confident, but I realized that I was capable of contributing more than I thought. As I gained more experience working with my team and on the project, I became more comfortable in both technical work and group discussions. This showed me that stepping out of my comfort zone is necessary for growth.

Going forward, I want to apply this mindset in other classes and future internships. I plan to be more proactive, ask questions, and get involved even if I feel unsure at first. I also want to keep improving my communication skills, especially when explaining technical ideas. Overall, this experience helped me grow both personally and professionally and gave me more confidence in my abilities.

Academic Enhancement

This semester in EPICS helped me better understand what the engineering design process actually looks like in real situations. Working on the Bubble Tube project showed me that things don't always work the first time, and a lot of progress comes from trial and error. One of the main challenges we dealt with was sealing the tube to stop leaks. Over several weeks, we tried different methods like tape, Mega Loc, silicone, and epoxy. Each time, we tested the tube, saw what worked or didn't, and made changes. At first, it was a bit frustrating that nothing fixed the problem right away, but over time I realized that this is just part of the process. We weren't failing, we were learning what didn't work so we could improve the next attempt and move closer to a working solution.

This connects to what I've learned about engineering in general, especially the idea of iterations. In classes, problems usually have a clear answer, but in this project, there wasn't one right solution.

Instead, we had to keep testing and adjusting until we got closer to what worked. I also learned how important teamwork is in the design process. Everyone brought different ideas, and working together helped us figure out better solutions. Another thing I started to think about more was user experience. Since this project is meant for children, we had to consider things like safety, durability, and how the final product would actually be used.

In the future, I'll take this mindset into other classes and projects. I now understand that it's okay if something doesn't work right away, and that improvement comes from testing and making changes. I also plan to be more patient when working through problems and to stay open to trying different approaches. Overall, this experience helped me connect what I've learned in class to real world situations and gave me a better understanding of how engineering actually works.